

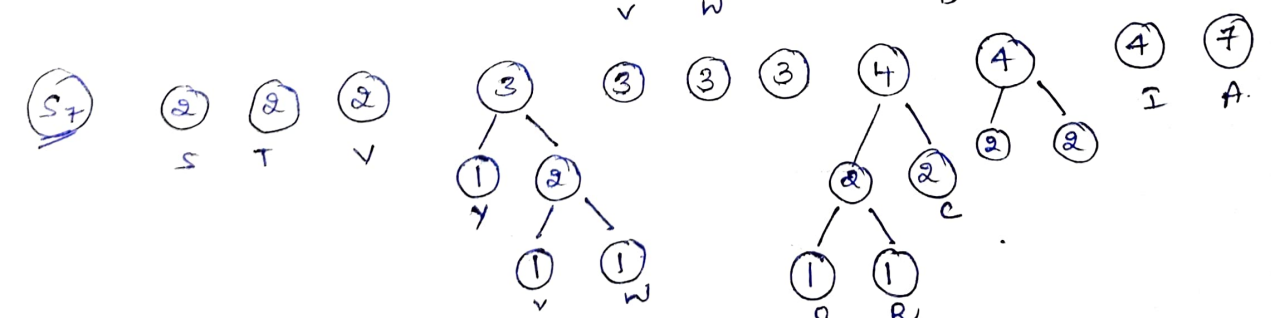
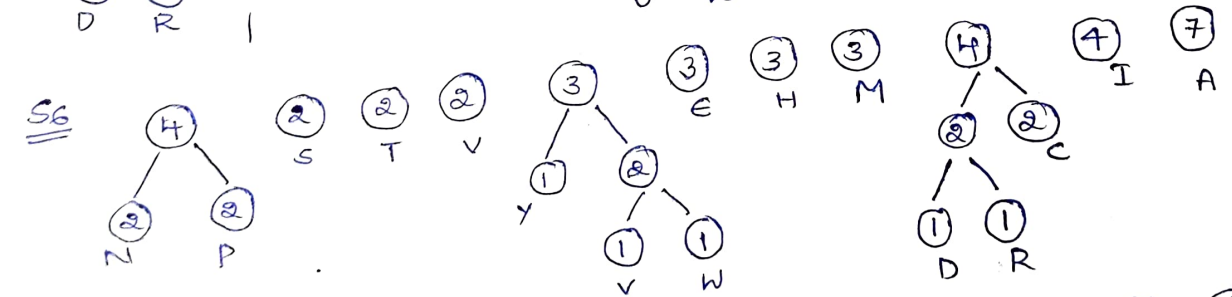
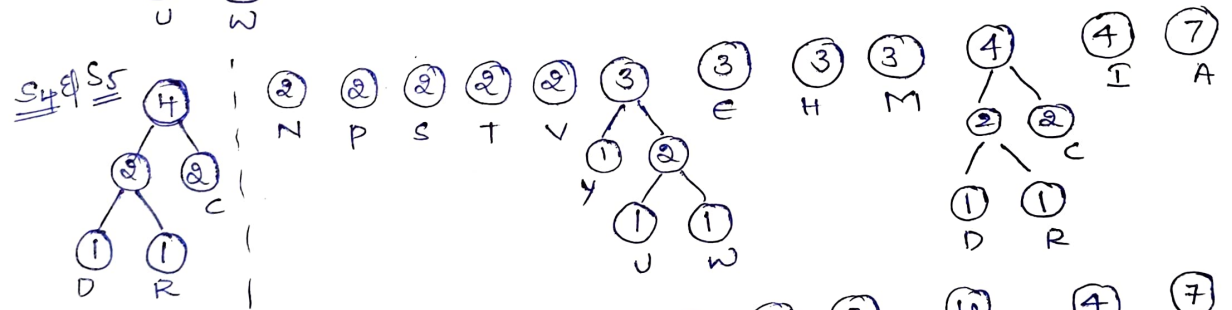
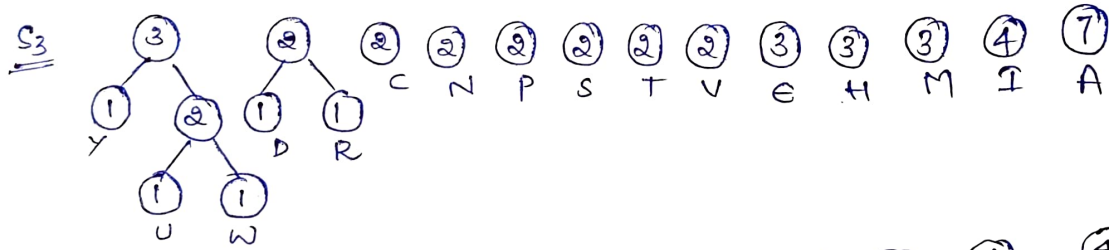
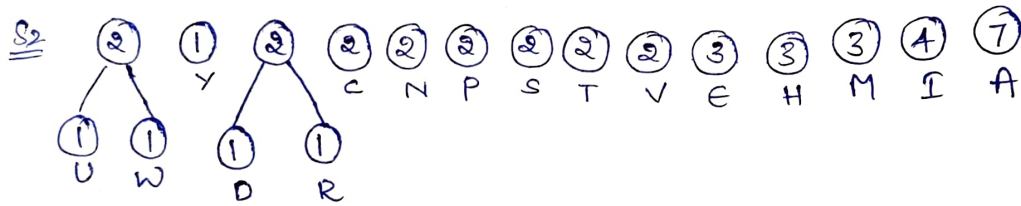
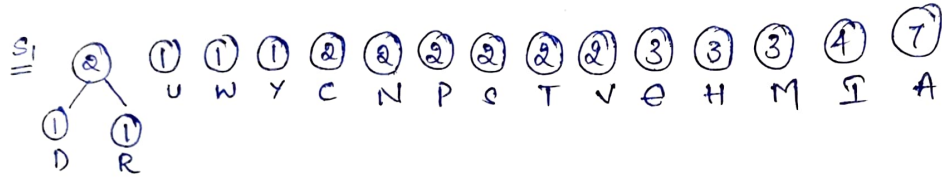
Q Using Huffman Coding arrive max encoding bits for phase

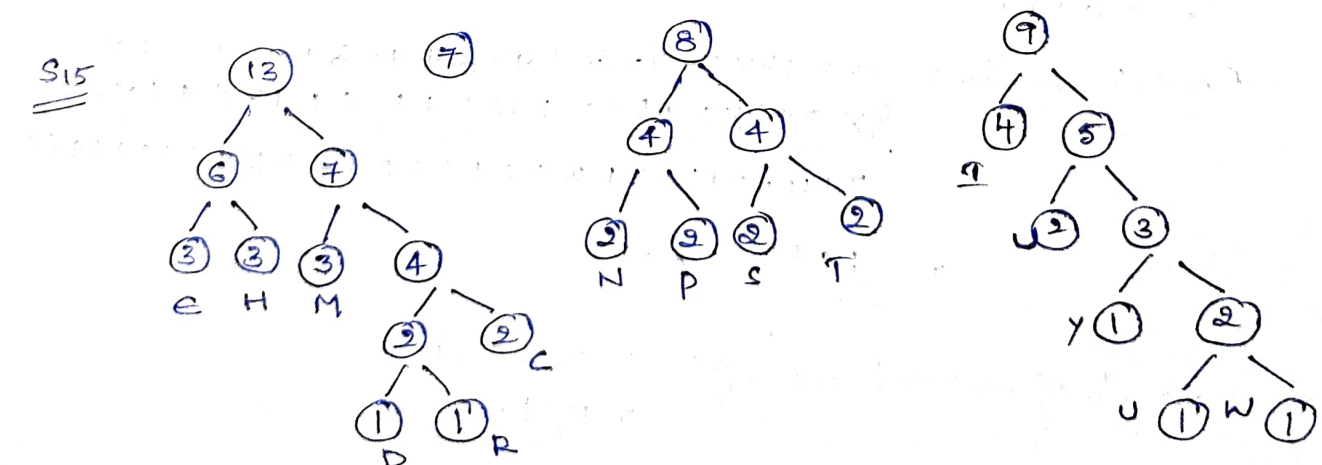
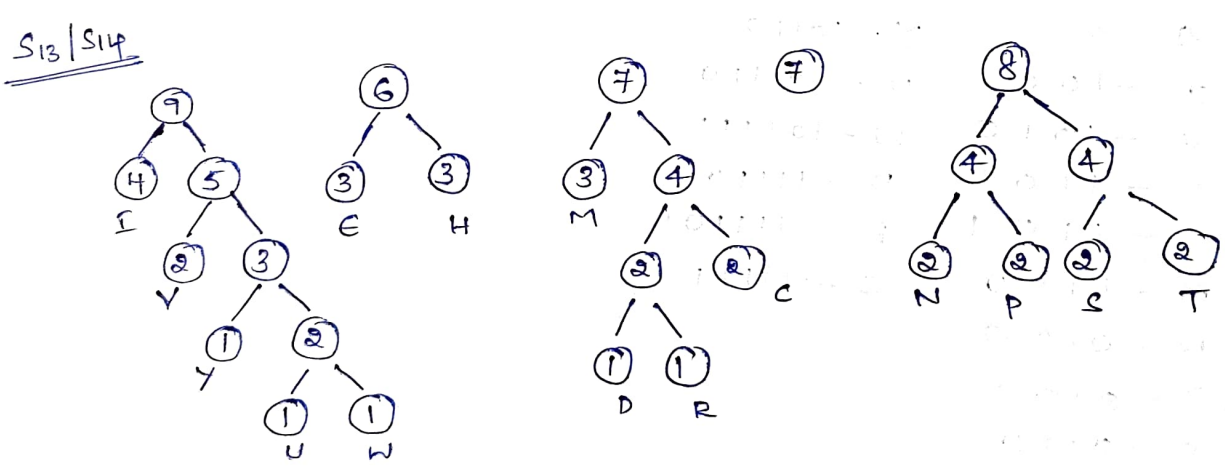
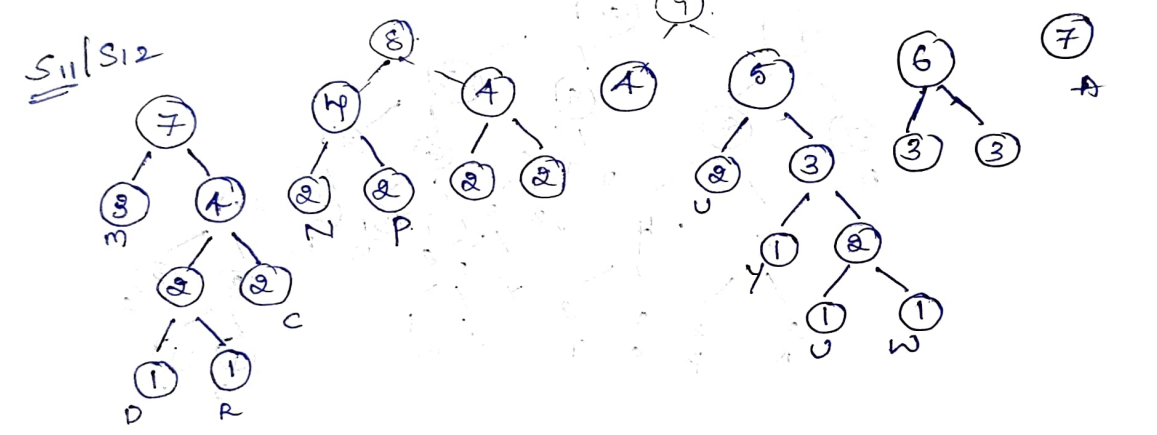
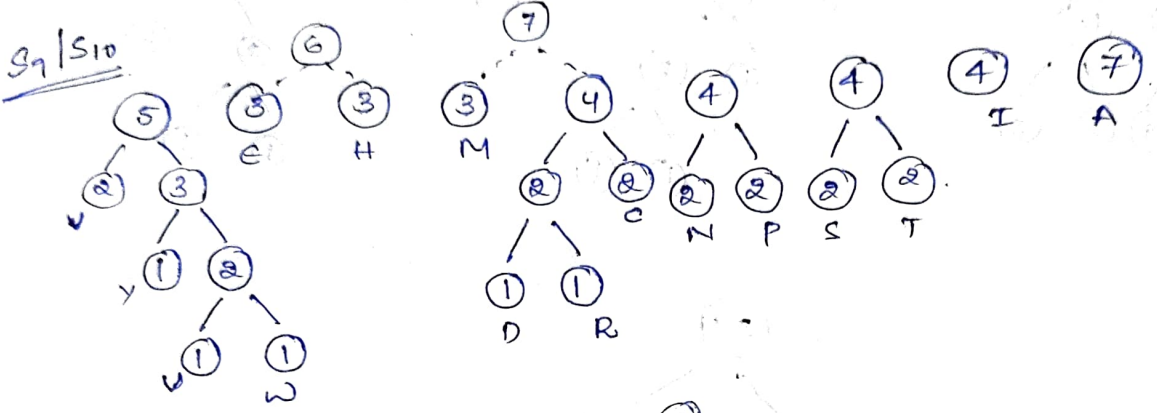
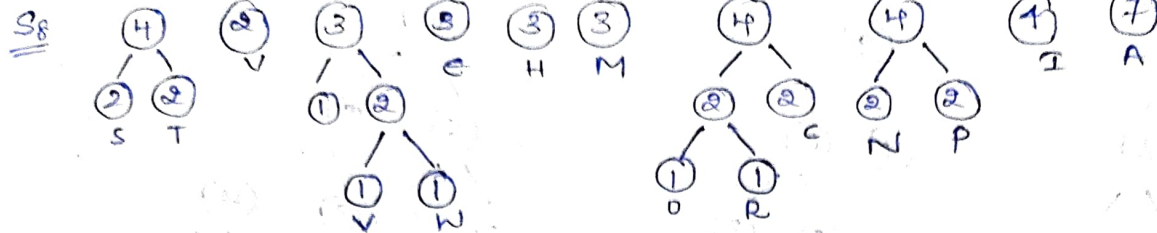
AMRITA VISHWA VIDYAPEETHAM CHENNAI CAMPUS.

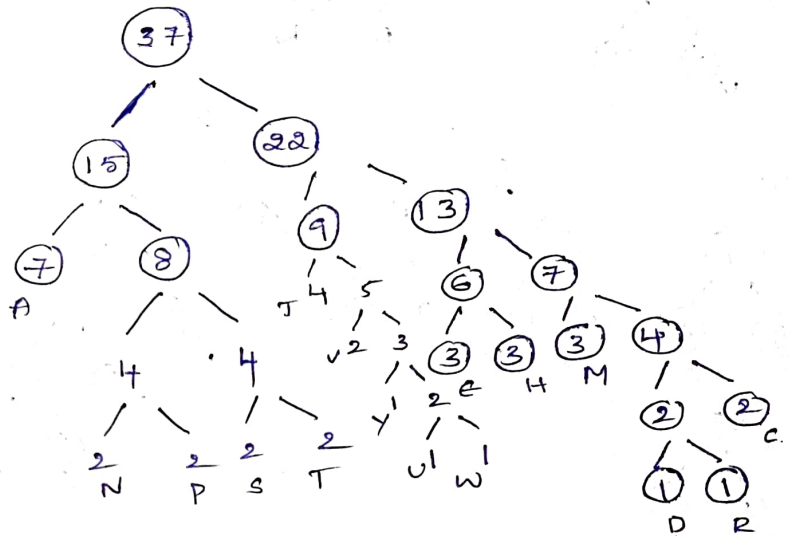
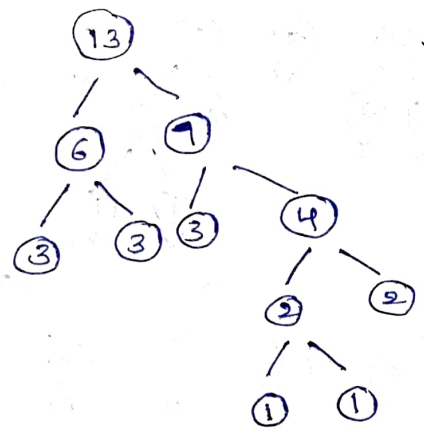
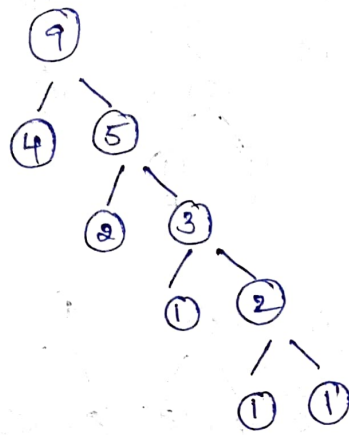
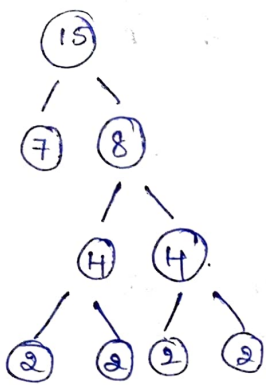
A M R I T A V I S H W D Y P E C N U
7 3 1 4 2 2 2 3 1 1 1 2 3 2 2 1

Arrange in ascending order.

① ① ① ① ① ② ② ② ② ② ② ③ ③ ③ ④ ⑦
D R U W Y C N P S T V E H M I A







A - 00

Y - 10110

I - 100

U - 101110

V - 1010

W - 101111

E - 1100

D - 111100

H - 1101

R - 111101

M - 1110

C - 11111

N - 0100

P - 0101

S - 0110

T - 0111

$$\begin{aligned} \text{Avg code length} &= \frac{7 \times 2 + 4 \times 3 + 4 \times 3 + 4 \times 3 + 4 \times 3 + 4 \times 2 + 4 \times 2 + 4 \times 2 + 4 \times 2 + 5 \times 2 + 1 \times 5 + 1 \times 6 + 1 \times 6 + 1 \times 6 + 1 \times 6}{7 + 4 + 3 + 3 + 3 + 2 + 2 + 2 + 2 + 2 + 2 + 1 + 1 + 1 + 1} \\ &= \frac{141}{37} = 3.81 \end{aligned}$$

length of encoded message $\Rightarrow 37 \times 3.81$
 $\Rightarrow 141$

2) Job scheduling.

Let there be 14 jobs for sequencing whose profits are
 $36, 21, 28, 19, 21, 13, 28, 25, 18, 20, 27, 22, 14, 26$.

Times are $5, 8, 3, 4, 4, 9, 12, 14, 2, 7, 5, 1, 6, 3$.

D.O $P \{36, 28, 28, 27, 26, 25, 22, 21, 21, 20, 19, 18, 14, 13\}$.

$T \{5, 3, 12, 5, 3, 14, 1, 8, 4, 7, 4, 2, 6, 9\}$

J_{12}	J_9	J_3	J_5	J_1	J_{13}	J_{10}	J_2	J_6				J_7		J_8
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14